

Condolence Letter & Terminal Hospitalization Summary

To: Dr. K. Li (Primary Care)

From: Hematology/Oncology Service

Date: 05.08.2023

Patient: Martha Sanders, MRN AML078

Primary diagnosis: Acute Myeloid Leukemia, **ELN 2022 Adverse, monosomal karyotype**

First-line therapy: Venetoclax + Decitabine (start 21.09.2022)

Notification of Death

It is with sadness that we inform you that **Ms. Martha Sanders (06.02.1953)** died on **21.07.2023** on our hematology service. The immediate cause of death was **septic shock due to carbapenem-resistant *Klebsiella pneumoniae* bacteremia**, occurring in the setting of **refractory AML**, profound chemotherapy-associated **neutropenia**, and **diffuse alveolar hemorrhage**. Her code status at the time of death was **DNR/DNI**, established after multiple goals-of-care meetings with Ms. Sanders and her family.

Diagnostic Background

- **Initial presentation (13.09.2022):** Two months of fatigue, gingival bleeding, and easy bruising. Admission labs: **WBC $13.2 \times 10^9/L$** (blasts 22%), **Hgb 7.4 g/dL**, **Plt $38 \times 10^9/L$** , **LDH 690 U/L**, uric acid 6.7 mg/dL.
 - **Bone marrow (13.09.2022):** 70–75% blasts; flow **CD34⁺/CD117⁺/HLA-DR⁺/CD13⁺/CD33⁺** with aberrant **CD7**.
 - **Cytogenetics:** Complex **monosomal karyotype** with -5, -7, del(17p), +8 and additional abnormalities (full ISCN in chart).
 - **Molecular:** TP53 frameshift (VAF 42%) and TET2 (VAF 18%); **FLT3/NPM1/IDH1/2 negative**.
 - **Baseline echo:** LVEF **58%**. Hepatitis B/C and HIV negative.
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Treatment Course & Response

First-line regimen: Decitabine 20 mg/m² D1–5 q28d + Venetoclax (ramp to 400 mg daily D1–21), **began 21.09.2022**.

- **Cycle 1–2 (Sep–Nov 2022):** Predictable cytopenias; short **febrile neutropenia** (cultures negative) responded to piperacillin–tazobactam. No TLS.
- **Marrow 28.11.2022:** Partial cyto-reduction (~**12% blasts**), not a complete remission; venetoclax continued with careful antimicrobial management (mold-active azole **avoided** to preserve dosing; trialed **miconazole** during deepest nadirs).
- **Winter 2022/23:** Intermittent RBC/platelet support; ECOG 1–2; ongoing outpatient.
- **Clinical progression documented 21.04.2023:** rising LDH (980 U/L), circulating blasts 6–10%, marrow ~**28% blasts** with persistent TP53 mutation; transfusion needs increasing. After shared decision-making, we discussed options (CPX-351, FLAG-IDA, palliative venetoclax continuation, clinical trials). She **declined intensive inpatient salvage** and **was ineligible** for TP53-directed trials due to renal function and performance status.

Supportive & palliative phase (late April–July 2023):

- Continued **lower-intensity venetoclax (200 mg D1–14)** with **decitabine D1–5** as tolerated, primarily for symptom control; frequent breaks for infections/cytopenias.
- Antimicrobial prophylaxis: **levofloxacin** and **acyclovir** routinely; **miconazole** during prolonged neutropenia.
- Transfusion thresholds: Hgb ≥ 8 g/dL; platelets $\geq 10 \times 10^9$ /L (≥ 20 with bleeding).
- Palliative care integrated; advance directive completed naming her daughter, **Olivia Sanders**, as surrogate.

Final Admission (15.07.2023 → 21.07.2023)

Reason for admission: Fever, rigors, pleuritic chest pain, and hemoptysis.

On arrival: T 38.9 °C, BP 86/54 (MAP 61), HR 124, RR 28, SpO₂ 90% RA → 95% on 6 L NC. ECOG 3.

Key labs (15.07): WBC 0.6×10^9 /L (ANC 0.02), Hgb 6.8 g/dL, Plt 7×10^9 /L; Cr 1.8 mg/dL (baseline 1.2), AST/ALT 58/64, Tbili 1.4; **lactate 4.6 mmol/L; fibrinogen 160 mg/dL, INR 1.6, D-dimer >10,000 ng/mL FEU.**

Imaging: CT chest — bilateral patchy consolidations with ground-glass opacities; small pleural effusions.

Microbiology:

- **Blood cultures** (central + peripheral) positive at 10 hours for **Klebsiella pneumoniae** with carbapenemase gene detected (NDM).
- **Respiratory PCR**: negative for influenza/RSV/COVID; serum **galactomannan 0.4** (indeterminate).

Management:

- Broad-spectrum antibiotics **meropenem** → **ceftazidime/avibactam + aztreonam** per ID.
- **Micafungin** initiated; **acyclovir** continued.
- Platelets and cryoprecipitate for DAH and hypofibrinogenemia; RBCs to maintain Hgb ≥ 8 ; **tranexamic acid nebulization** trialed.
- High-flow nasal oxygen; cautious diuresis; electrolyte repletion (K >4 , Mg >2).
- Despite therapy, she developed **refractory shock, worsening hemoptysis**, and **acute kidney injury** (Cr 2.4). After discussion with Ms. Sanders and family, life-prolonging measures were limited to comfort-oriented care and noninvasive oxygen.

Time of death: 21.07.2023 at 05:42. Family at bedside; chaplain present. Death certificate completed.

Comorbidities

- **Stage 3a CKD, hypertension** (lisinopril 10 mg daily), **hypothyroidism** (levothyroxine 75 μ g), **hyperlipidemia** (pravastatin 20 mg), **osteoarthritis** (acetaminophen only), remote **tobacco** (quit 2005).
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Medications Near End of Life

Ceftazidime/avibactam + aztreonam, micafungin, acyclovir, hydromorphone infusion for dyspnea/pain, lorazepam PRN, pantoprazole, ondansetron, bowel regimen, frequent platelet/cryoprecipitate and PRBC transfusions. Home lisinopril/statin held.

Aftercare

Next of kin: **Olivia Sanders** (daughter). Bereavement resources provided. Autopsy declined. We extend our condolences and remain available for any documentation needs.

Sincerely,
[Name], MD — Consultant Hematologist